Alireza Nasirzadeh, PhD

Research Professor · Assistive and Rehabilitation Robotics Laboratory

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PROFESSIONAL EXPERIENCE

Research Professor Feb 2022 – Present
Assistive and Rehabilitation Robotics Laboratory, School of Mechanical Engineering
Chung-Ang University, South Korea

Editorial Board Member 2026 – Present
Journal of NeuroEngineering and Rehabilitation

Biomechanist Jan 2020 – Oct 2021
Athlete Evaluation Department, I.R.I. National Paralympic Committee, Iran

Assistant Professor Sep 2019 – Nov 2021
Faculty of Physical Education and Sport Sciences, Kharazmi University, Iran

Teaching Assistant Jan 2016 – May 2019
Various Universities, Iran

EDUCATION

PhD in Sports Biomechanics Sep 2012 – Sep 2016
Kharazmi University, Iran
Thesis: Comparison of Selected Kinematic, Kinetic, and Functional Lower Limb Variables during Walking and Running at Different Speeds among Young Male Athletes Focusing on Limb Dominance

MSc in Physical Education and Sports Sciences Sep 2009 – May 2011
University of Birjand, Iran
Dissertation: Architectural Characteristics of the Upper Extremity Skeletal Muscles in Young Male Table Tennis Players

BSc in Mathematics Sep 2003 – May 2009
University of Birjand, Iran

RESEARCH AREAS

Movement Biomechanics · Wearable Robotics & Exosuits · Musculoskeletal Modeling · Dynamic Stability and Balance · Rehabilitation Engineering · Injury Mechanics

PUBLICATIONS

Nasirzadeh, A. et al. (2026). Biomechanical Effects of Passive Exosuit Assistance on Tibiofemoral Loading and Dynamic Stability during Downhill Walking. *Journal of NeuroEngineering and Rehabilitation.*

Park, C., **Nasirzadeh, A.**, Lee, K., Kim, G., & Lee, G. (2026). Real-Time Estimation of User Adaptation During Hip Exosuit-Assisted Walking Using Wearable Inertial Measurement Unit Data and Long Short-Term Memory Modeling. *Biomimetics,* 11(2), 96.

Nasirzadeh, A. et al. (2025). Harnessing Resistance: The Impact of a Passive Exosuit on Golf Swing Biomechanics. *Journal of Biomechanics,* 192, 112958.

Choe, E., Moon, J., Ryu, J., Yang, S., **Nasirzadeh, A.**, Kong, S., Choi, Y., & Lee, G. (2025). Low-Profile, Shoe-Type Ankle-Foot Orthosis with Active Variable Ankle Stiffness via Wire-Fabric Compression Mechanism. *Biomimetics,* 10(8), 539.

Kim, Y., Park, S., Yang, S., **Nasirzadeh, A.**, Lee, G. (2024). The Interplay Between Muscular Activity and Pattern Recognition of Electro-Stimulated Haptic Cues During Normal Walking: A Pilot Study. *Bioengineering,* 11, 1248.

Nasirzadeh, A. et al. (2023). The Effects of a Custom-Designed High-Collar Shoe on Muscular Activity, Dynamic Stability, and Leg Stiffness: A Biomimetic Approach Study. *Biomimetics,* 8, 274.

Park, J., Nam, K., Yun, J., Moon, J., Ryu, J., Park, S., **Nasirzadeh, A.** et al. (2023). Effect of hip abduction assistance on metabolic cost and balance during human walking. *Science Robotics,* 8(83), eade0876.

Nasirzadeh, A. et al. (2023). Influence of circumferential ankle pressure of shoe collar on the kinematics, dynamic stability, electromyography, and plantar pressure during normal walking. *PLOS ONE,* 18(2), e0281684.

Nasirzadeh, A. (2020). The influence of horizontal velocity on inter-lower-limbs local and global asymmetry during walking. *Physical Treatments,* 10(2), 69-78.

Nasirzadeh, A., Mohseni, F. (2018). Local and global symmetry during jogging in young male athletes. *Physical Treatments,* 8(1), 17-26.

Nasirzadeh, A. et al. (2017). Gait symmetry and its evaluation methods: a review. *Scientific Journal of Rehabilitation Medicine,* 6(2), 283-97.

Nasirzadeh, A. et al. (2017). A review of selected factors that affect gait symmetry. *Physical Treatments,* 7(1), 3-12.

Nasirzadeh, A. et al. (2016). The influence of movement velocity on inter-lower-limbs kinematical symmetry in normal walking: a pilot study. *Scientific Journal of Rehabilitation Medicine,* 5(4), 159-72.

Nasirzadeh, A. et al. (2015). Multivariate analysis of 200-m front crawl swimming performance in young male swimmers. *Acta of Bioengineering and Biomechanics,* 17(3), 137-43.

Nasirzadeh, A. et al. (2014). Selected anthropometric, muscular architecture, and biomechanical variables as predictors of 50-m performance of front crawl swimming in young male swimmers. *Science and Sports,* 29(5), e75-e81.

Nasirzadeh, A. et al. (2014). Relationship between sprint performance of front crawl swimming and muscle fascicle length in young swimmers. *Journal of Sports Science and Medicine,* 13(3), 550-556.

UNDER REVIEW OR IN PROGRESS

Nasirzadeh, A. (in progress). Shank Attachment Geometry Governs Multi-Joint Force Transmission During Handle-Supported Walking. *IEEE Transactions on Neural Systems and Rehabilitation Engineering.*

Kim, G., Lee, S., **Nasirzadeh, A.** (under review). Rapid Hip-Abduction Exosuit-Induced Torque to Improve Mediolateral Standing Balance Without Stepping. *Nature Biomedical Engineering.*

Yang, J., **Nasirzadeh, A.** (under review). Mechanically Encoded Asymmetric Force Allocation for Single-Motor Bilateral Hip Assistance. *Advanced Intelligent Systems.*

PRESENTATIONS

PODIUM PRESENTATIONS

Mediolateral Balance Recovery during Walking using H-Medi Exosuit (2026). **44th International Society of Biomechanics in Sports Conference (ISBS)**, Loughborough, UK.

Design and Validation of a Passive Exosuit to Enhance Golf Swing Performance (2025). **7th International Conference on Medicine in Novel Technology and Devices**, Hangzhou, China.

Power Up the Swing: A Biomechanical Study Towards a Golf Exosuit (2024). **42nd International Society of Biomechanics in Sports Conference (ISBS)**, Salzburg, Austria.

Active Variable Compression Shoes: Design and Evaluation of an External Ankle Supporter to Prevent Ankle Sprain via Active Compression Mechanism (2024). **10th IEEE RAS/EMBS International Conference for Biomedical Robotics and Biomechatronics (BioRob)**, Heidelberg, Germany.

Isokinetic analysis of elite Iranian Paralympic athletes (2021). **Second National Olympic & Paralympic Congress**, Tehran, Iran.

Epidemiology of Sports Injuries and Injury Prevention Strategies in Para Powerlifting (2021). **Asian Paralympic Committee** (online).

The effect of long-term custom-made foot orthoses on foot structures in children with flexible flat feet (2018). **Eleventh International Conference of Sport Sciences**, Tehran, Iran.

The effect of gender and anthropometrical parameters on the dynamics of sprint start among young elite sprinters (2012). **XXX International Society of Biomechanics in Sports (ISBS) Conference**, Melbourne, Australia.

POSTER PRESENTATIONS

Reducing Tibiofemoral Joint Force During Downhill Walking Using Various Passive Exosuits: A Pilot Study (2025). **XXX Congress of the International Society of Biomechanics (ISB)**, Stockholm, Sweden.

Passive Collar Stiffness of a Custom-designed High-Collar Shoe (2023). **XXIX Congress of the International Society of Biomechanics (ISB)**, Fukuoka, Japan.

The effect of a researcher-made soft insole on selected kinematic and kinetic variables of walking of active girls (2015). **First National Conference on New Findings in Sport Sciences**, Tehran, Iran.

Progress in front crawl swimming speed performance in adolescent swimmers, emphasizing biomechanics, muscular architecture, and anthropometric factors (2013). **Second National Conference on Sports Talent Identification**, Tehran, Iran.

Relationship between sprint performance of front crawl swimming and muscle fascicle length in young swimmers (2012). **XXXII FMSI World Congress of Sports Medicine**, Rome, Italy.

Ultrasonographic examination of muscle architecture parameters of the upper arm muscles of table tennis players aged 40-60 years (2011). **Sixth National Conference on Physical Education and Sport Sciences**, Tehran, Iran.

Investigating the indicators of muscular architecture in different parts of the brachial muscles of table tennis players (2011). **First International Sports Medicine Congress**, Isfahan, Iran.

TEACHING EXPERIENCE

Instructor · Kharazmi University 2016-2021
Sports Biomechanics, Research Methods in Biomechanics, Statistics in Sports and Exercise, Anatomy and Kinesiology, Advanced Methods in Biomechanics, Principles of biomechanical data collection and analysis.

Workshops

Physical Fitness Parameters in Children with Disabilities (2018). **I.R.I. Sports Federation for the Disabled**.

Principles of Motion Analysis and Interpreting Results using Visual3D Software (2018). **Kharazmi University**.

Force Plate: Applications in Sport Sciences and Data Analysis (2017). **Shiraz University**.

Biomechanical Assessment of Athletic Performance (2014). **I.R.I. Sports Federation for the Disabled**.

AWARDS & HONORS

Outstanding Student 2016
Department of Sports Biomechanics, Kharazmi University

Outstanding Presentation Award in Biomechanics 2015
First National Conference on New Findings in Sport Sciences

New Investigator Award 2011
Sixth National Conference on Physical Education and Sport Sciences

SERVICE & OUTREACH

Editorial Roles
Editorial Board Member, Journal of NeuroEngineering and Rehabilitation

Ad-Hoc Reviewer
Journal of NeuroEngineering and Rehabilitation, Journal of Sports Sciences, Sports Biomechanics, International Journal of Athletic Therapy & Training, BMC Musculoskeletal Disorders, PLOS ONE, Scientific Reports, IEEE Transactions on Neural Systems & Rehabilitation Engineering.

MEMBERSHIPS & AFFILIATIONS

International Society of Biomechanics
International Society of Biomechanics in Sports

TECHNICAL SKILLS

- **Biomechanical Analysis:** Gait, kinematics, kinetics, EMG
- **Wearable Robotics:** Exosuits and rehabilitation devices
- **Musculoskeletal Modeling:** AnyBody Modeling System, Visual3D
- **Programming:** MATLAB, Python, LabVIEW
- **Experimental Systems:** Vicon, Qualisys, force plates, IMUs

ENGLISH LANGUAGE SKILLS

PTE Academic 2024
Overall, 81/90 - Listening 85, Reading 79, Writing 80, Speaking 90

GRE (General) 2021
Verbal Reasoning 140, Quantitative Reasoning 163 (out of 170)

IELTS Academic 2021
Overall, 7.5/9 (CEFR C1) - Listening 8.5, Reading 7.5, Writing 6.5, Speaking 7.0

LICENSES & CERTIFICATES

Coaching Certification in Physical Fitness for Individuals with Disabilities 2019
Judo Coaching Certificate 2018
Judo Black Belt, 1st Dan 2015
Swimming Pool Lifeguard Certificate 2011
Table Tennis Coaching Certificate 2009

REFERENCES

Giuk Lee
Associate Professor, Department of Mechanical Engineering, Chung-Ang University
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Yong Ung Kwon
Associate Professor, College of Sport Sciences, Chung-Ang University
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Saeed Ilbeigi
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